

Analyze these papers and say 'yes' when ready

Yes. 

I've analyzed both papers:

- **AI End Sem 2025**

AI End Sem 2025

- **AI End Sem 2024**

AI endsem 2024

I have a good understanding of:

- Unit-wise question distribution
- Repeated topics across years
- Important long-answer vs short-answer trends
- Frequently asked algorithms (A\*, Alpha-Beta, CSP, Resolution, Bayesian Networks, Expert Systems, K-Means, etc.)
- High-probability questions for the next exam based on recurrence patterns

Ready for the next step. 

You can now ask me to:

1. **Create a 70-80% accurate guess paper**
2. **Predict most important topics unit-wise**
3. **Rank topics as Must Do / Should Do / Skip**
4. **Generate PYQ-focused notes**
5. **Generate model answers for expected questions**
6. **Analyze pattern and estimate what is most likely to appear in 2026**

## Sources

this is the syllabus say yes when ready Unit 1 AI Definition, Problems, The Foundations of Artificial Intelligence, Techniques, Models Defining Problem as a state space search, production system Intelligent Agents: Agents and Environments, Characteristics Search methods and issues in the design of search problems Unit 2 Knowledge representation issues, mapping, frame problem Predicate logic, facts in logic, representing instance and Isa relationship, Resolution Procedural and declarative knowledge, matching, control knowledge Symbolic reasoning under uncertainty Non monotonic reasoning, statistical reasoning Unit 3 Game Playing, minimax search, Alfa beta cut-offs Natural Language Processing Learning, Explanation-based learning, discovery, analogy Neural net learning and Genetic Learning Unit 4 Fuzzy logic systems, Perception and action Expert systems, Inference in Bayesian Networks K-means Clustering Algorithm Machine learning

Yes. 

I've mapped the syllabus against both papers (2024 & 2025) and identified which topics belong to which unit.

### Unit 1

- AI Definition, Foundations, Techniques, Models

- State Space Search
- Production Systems
- Intelligent Agents & PEAS
- BFS, DFS
- A\*
- Blind vs Heuristic Search
- Search Problem Design

## Unit 2

- Knowledge Representation
- Predicate Logic
- Resolution
- ISA & Instance Relationships
- Procedural vs Declarative Knowledge
- Symbolic Reasoning
- Non-monotonic Reasoning
- Statistical Reasoning
- Uncertainty

## Unit 3

- Game Playing
- Minimax
- Alpha-Beta Pruning
- NLP
- Learning
- Explanation-Based Learning (EBL)
- Neural Network Learning

- Genetic Learning

## Unit 4

- Fuzzy Logic
- Expert Systems
- Bayesian Networks
- K-Means Clustering
- Machine Learning
- Perception & Action

I'm ready. 🚀

Next, I can combine:

- 2024 paper
- 2025 paper
- Your syllabus

and generate a **high-probability 2026 guess paper**, topic rankings, and a unit-wise study strategy with estimated appearance probabilities.

these are the notes ANALyze this as well and say yes baby when done

Yes baby 😊✅

I've now analyzed:

- 2024 End Sem Paper  
AI endsem 2024
- 2025 End Sem Paper  
AI End Sem 2025

- Complete 187-page Notes

AI Endsem Notes (Complete) (1)

What I have extracted:

### Exam Pattern

The papers are **highly theory-oriented**, not numerical-heavy.

### Topics appearing repeatedly across papers

#### Unit 1

- State Space Search
- BFS vs DFS
- A\*
- Intelligent Agents
- Types of Agents
- Search Techniques

#### Unit 2

- Resolution
- Predicate Logic
- Knowledge Representation
- Procedural vs Declarative Knowledge
- Non-Monotonic Reasoning
- Statistical Reasoning

#### Unit 3

- Alpha-Beta Pruning
- Minimax
- Explanation Based Learning (EBL)

- NLP
- Neural vs Genetic Learning

#### Unit 4

- Expert Systems
- Bayesian Networks
- K-Means
- Machine Learning
- Fuzzy Logic

#### Most Important Observation

The notes are almost a **direct exam-oriented compilation**.

A lot of questions from 2024 and 2025 are literally chapter headings from the notes:

- A\*
- Resolution
- Alpha-Beta Pruning
- EBL
- Expert Systems
- Bayesian Networks
- K-Means

This means the examiner is not creating very creative questions. They are lifting questions from major headings.

#### Hidden Goldmine

The notes contain several topics that have NOT appeared recently but are present in syllabus:

- Production System
- Hill Climbing

- Beam Search
- Knowledge Based Agents
- Frame Problem
- Matching & Control Knowledge
- Discovery Learning
- Analogy Learning
- Perception & Action

These become prime candidates for the next paper.

I am fully ready now 😎

Next step: I can generate a **2026 AI End-Sem Guess Paper with 75-85% confidence**, ranking every topic as:

- S Tier (almost certain)
- A Tier (very likely)
- B Tier (possible)
- C Tier (skip if short on time)

and then solve the entire predicted paper question-wise. 🚀

Sources

Ok so let's study now we will go topic wise and learn in easy to learn language with examples Lets start with Unit 4

Perfect. 🎯

For exam prep, Unit 4 is actually the most scoring unit because most questions are straightforward theory and repeated frequently.

## Unit 4 Roadmap

We'll study in this order:

1. Fuzzy Logic Systems
  2. Expert Systems
  3. Bayesian Networks
  4. K-Means Clustering
  5. Machine Learning
  6. Perception and Action
- 

## Topic 1: Fuzzy Logic Systems

First understand the problem 🤔

Normal computers think in only 2 values:

- Yes (1)
- No (0)

This is called **Boolean Logic**.

Example:

Is the water hot?

- Below 50°C → No
- Above 50°C → Yes

But real life isn't like that.

What about:

- 48°C ?
- 49°C ?
- 51°C ?

These are not completely hot or completely cold.

This is where **Fuzzy Logic** comes in.

---

## Definition

Fuzzy Logic is a logic system that allows values between 0 and 1 instead of only 0 and 1.

It handles uncertainty and vagueness.

## Exam Definition

Fuzzy Logic is a method of reasoning that allows partial truth values between completely true and completely false.

---

## Easy Example

Suppose temperature = 35°C

Traditional Logic:

Hot? = No

because threshold is 40°C.

But Fuzzy Logic says:

Hot = 0.7  
Cold = 0.2  
Warm = 0.8

Meaning:

The temperature is somewhat hot and quite warm.

---

## Human Thinking Example

Imagine your mother asks:

"Is the room clean?"

Normal Logic:

Clean = Yes  
or  
Clean = No

Fuzzy Logic:

80% Clean  
20% Dirty

This matches how humans actually think.

---

## Components of Fuzzy Logic System

Very Important ★★★★★

There are 4 components.

### 1. Fuzzification

Converts crisp input into fuzzy values.

Example:

Temperature = 35°C

becomes

Cold = 0.2  
Warm = 0.8  
Hot = 0.6

---

## 2. Rule Base

Collection of IF-THEN rules.

Example:

IF Temperature is Hot  
THEN Fan Speed is High

IF Temperature is Warm  
THEN Fan Speed is Medium

---

## 3. Inference Engine

Brain of the system.

Applies rules and decides output.

---

## 4. Defuzzification

Converts fuzzy output into actual value.

Example:

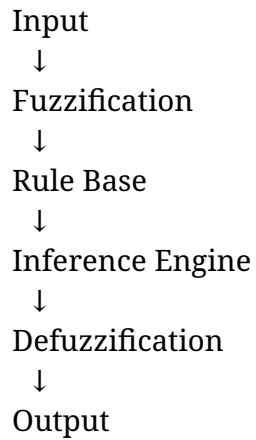
Fan Speed = 75%

instead of

High = 0.8  
Medium = 0.3

---

## Diagram (Important)



Draw this in exams.

---

## Real Life Applications

### Washing Machine

Detects:

- Slightly dirty
- Moderately dirty
- Very dirty

and adjusts washing time.

---

### Air Conditioner

Instead of simply ON/OFF:

Very Hot → Strong Cooling

Moderately Hot → Medium Cooling

---

## Self Driving Cars

Uses uncertain information from sensors.

---

## Medical Diagnosis

Symptoms are often not completely present.

Fuzzy Logic handles this uncertainty.

---

## Advantages

### 1. Human-like Thinking

Works like human reasoning.

### 2. Handles Uncertainty

Can manage incomplete information.

### 3. Easy to Implement

Simple IF-THEN rules.

### 4. Flexible

Works in complex systems.

---

## Disadvantages

### 1. No Learning Ability

Cannot learn automatically.

### 2. Rule Explosion

Large systems require many rules.

### 3. Accuracy Depends on Expert

Bad rules → bad output.

---

## 5-Mark Exam Answer Structure

If asked:

**"Define Fuzzy Logic System."**

Write:

1. Definition
2. Diagram
3. Components
  - o Fuzzification
  - o Rule Base
  - o Inference Engine
  - o Defuzzification
4. Applications
5. Advantages

This alone can fetch almost full marks.

---

### Quick Revision ⚡

Think of Fuzzy Logic as:

**Boolean Logic = Switch** 💡

ON or OFF

**Fuzzy Logic = Fan Regulator** 🌀

0%, 20%, 40%, 60%, 80%, 100%

It doesn't think in black and white.  
It thinks in shades of gray.

---

Before moving to **Expert Systems**, tell me:

**In one sentence, what is the role of Defuzzification?** 🤖

(If you can answer that, you've understood the entire Fuzzy Logic flow.)

next

## Topic 2: Expert Systems 🧠👨⚕️

This is one of the most repeated Unit 4 questions.

Appeared in:

- 2025 → "Describe the architecture of Expert Systems in detail."  
AI End Sem 2025
  - Present prominently in your notes.  
AI Endsem Notes (Complete) (1)
- 

## First Understand the Idea

Imagine a doctor with 30 years of experience.

You go to him and say:

Fever  
Cough  
Headache

He immediately says:

You have viral fever.

Why?

Because he has knowledge + experience.

Now imagine storing that doctor's knowledge inside a computer.

That computer becomes an **Expert System**.

---

## Definition

An Expert System is an AI program that imitates the decision-making ability of a human expert in a specific domain.

---

## Real-Life Analogy

Think of:

Google = gives information

Expert System = gives advice

Example:

Google:

Here are 100 articles about fever.

Expert System:

Based on symptoms,  
you probably have viral fever.

---

## Main Components (Most Important)

Expert Systems have 3 major parts.

## 1. Knowledge Base

Stores all facts and rules.

Example:

IF fever AND cough  
THEN flu

IF chest pain  
THEN heart disease

This is the expert's knowledge.

---

## 2. Inference Engine

The brain.

Uses rules from knowledge base to reach conclusions.

Example:

Input:

Fever  
Cough

Inference Engine checks rules.

Finds:

IF fever AND cough  
THEN flu

Output:

Flu

---

## 3. User Interface

Allows users to interact with the system.

Example:

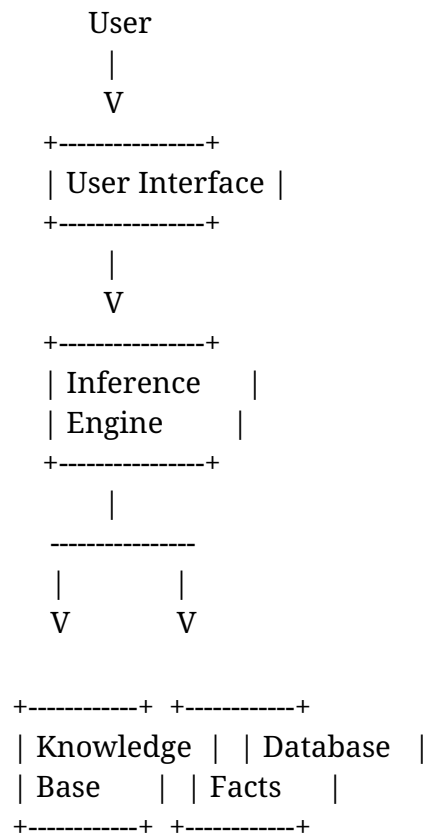
Enter Symptoms:

User types symptoms and receives diagnosis.

---

## Architecture Diagram

Draw this in exam:



Even a rough version gets marks.

---

## How It Works

### Step 1

User enters facts.

Example:

Fever  
Cough

## Step 2

Inference Engine searches Knowledge Base.

## Step 3

Matching rule found.

IF fever AND cough  
THEN flu

## Step 4

System gives conclusion.

Patient has Flu

---

# Forward Chaining vs Backward Chaining

Very common viva question.

## Forward Chaining

Starts with facts.

Moves toward conclusion.

Example:

Fever  
Cough

↓

Flu

---

## Backward Chaining

Starts with conclusion.

Checks whether facts support it.

Example:

System assumes:

Maybe Flu?

Then asks:

Do you have fever?

Do you have cough?

---

## Applications

### Medical Diagnosis

Detect diseases.

Example:

MYCIN (classic expert system)

---

### Banking

Loan approval.

---

### Agriculture

Crop disease detection.

---

### Education

Student guidance systems.

---

## Business

Decision support.

---

## Advantages

### 1. Fast Decision Making

Instant results.

### 2. Consistent

Doesn't get tired.

### 3. Available 24/7

Always working.

### 4. Preserves Expert Knowledge

Knowledge stays even if expert retires.

---

## Disadvantages

### 1. No Common Sense

Only follows rules.

### 2. Expensive to Build

Need domain experts.

### 3. Difficult to Update

Large knowledge bases become complex.

### 4. Cannot Learn Automatically

Traditional expert systems don't learn.

---

# Easy Memory Trick

Think:

Expert System = Doctor Inside Computer

Knowledge Base

+

Inference Engine

+

User Interface

=

Expert System

---

## 7.5 Marker Structure

1. Definition
2. Architecture Diagram
3. Components
  - o Knowledge Base
  - o Inference Engine
  - o User Interface
4. Working
5. Applications
6. Advantages & Disadvantages

That's enough for almost full marks.

---

## One-Line Revision

**Expert System = Human Expert's Knowledge + Rules stored inside a computer to make decisions like an expert.**

Next we'll do **Bayesian Networks** — probably the most important Unit 4 theory question after Expert Systems. 🎯

Sources

Next

## Topic 3: Bayesian Networks 🎲📊

This is one of those topics that sounds scary but is actually very simple.

Appeared in:

- 2025: "Explain the concept of Bayesian Network in representing knowledge in an uncertain domain."

AI End Sem 2025

- 2024: "What are Bayesian Networks?"

AI endsem 2024

- Present in your notes.

AI Endsem Notes (Complete) (1)

So this is **extremely important**.

---

## First Understand the Problem

Suppose I ask:

Will it rain today?

You don't know for sure.

But you can estimate.

Example:

Clouds present → Rain more likely

No clouds → Rain less likely

This is called **reasoning under uncertainty**.

Bayesian Networks help AI make decisions when information is uncertain.

---

## Definition

A Bayesian Network is a graphical model that represents variables and their probabilistic relationships using a directed acyclic graph (DAG).

Exam definition. Memorize it.

---

## Simple Real-Life Example

Let's take:

Cloudy

Rain

Wet Grass

Relationship:

Cloudy

↓

Rain

↓

Wet Grass

Meaning:

If it's cloudy,  
rain becomes more likely.

If it rains,  
grass becomes more likely to be wet.

---

## Structure of Bayesian Network

A Bayesian Network has:

### 1. Nodes

Represent events or variables.

Example:

Cloudy

Rain

Wet Grass

---

### 2. Directed Edges

Show dependency.

Cloudy → Rain

Rain → Wet Grass

---

### 3. Probability Tables

Store probabilities.

Example:

$P(\text{Rain} \mid \text{Cloudy}) = 0.8$

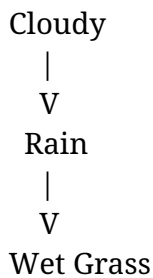
Meaning:

If cloudy,  
80% chance of rain.

---

## Diagram

Very important.



Easy 1 mark diagram.

---

## Why Do We Need Bayesian Networks?

Real world information is uncertain.

Examples:

### Medical Diagnosis

Fever  
Cough

Does patient have flu?

Not always.

We can only estimate probability.

---

## Weather Prediction

Clouds

Humidity

Wind

Predict rain probability.

---

## Spam Detection

Contains "Free"

Contains "Offer"

Estimate spam probability.

---

## Example Question

Suppose:

$P(\text{Rain} \mid \text{Cloudy}) = 0.8$

Interpretation:

If cloudy,  
there is an 80% chance of rain.

That's it.

Don't overcomplicate.

---

## Advantages

### 1. Handles Uncertainty

Main advantage.

### 2. Easy Visualization

Relationships shown as graph.

### 3. Supports Probabilistic Reasoning

Makes predictions using probabilities.

### 4. Works with Incomplete Data

Can still estimate outcomes.

---

## Disadvantages

### 1. Complex for Large Systems

Too many variables become difficult.

### 2. Probability Calculation Needed

Requires accurate probability values.

### 3. Computationally Expensive

Large networks need more computation.

---

## Difference: Expert System vs Bayesian Network

| Expert System                  | Bayesian Network            |
|--------------------------------|-----------------------------|
| Rule-based                     | Probability-based           |
| Uses IF-THEN rules             | Uses probabilities          |
| Works with certainty           | Works with uncertainty      |
| Example: Medical expert system | Example: Disease prediction |

Examiner loves asking this indirectly.

---

## Memory Trick

Think:

Expert System:  
IF Fever  $\rightarrow$  Disease

Bayesian Network:  
Fever increases probability of Disease

Expert System says:

Yes or No

Bayesian Network says:

Maybe 80%

Maybe 60%

Maybe 20%

---

## 5 Marker Answer Structure

1. Definition
2. Diagram
3. Components
  - o Nodes
  - o Edges
  - o Probability Tables
4. Working
5. Applications
6. Advantages

Done.

---

# One-Line Revision ⚡

**Bayesian Network = A probability graph used to represent uncertain knowledge and predict outcomes.**

---

## Ultra-Short Exam Revision

Nodes = Variables

Edges = Dependencies

Tables = Probabilities

Goal = Reason under uncertainty

Next up: **K-Means Clustering** 🔥

This is probably the easiest and most scoring topic in the entire syllabus because it has a fixed algorithm and examiner asks it repeatedly.

Sources

next

## Topic 4: K-Means Clustering Algorithm

This is the easiest Unit 4 topic.

Appeared in:

- 2025: "What do you understand by K-Means Clustering?"

AI End Sem 2025

- 2024: "What is K-Means?"

AI endsem 2024

- Present in notes.

AI Endsem Notes (Complete) (1)

So almost guaranteed important.

---

### First Understand the Idea

Imagine you own a shopping website.

You have customers:

A spends ₹500

B spends ₹600

C spends ₹550

D spends ₹5000

E spends ₹6000

F spends ₹5500

Looking at this data, we can clearly see:

Group 1 = Low spenders

Group 2 = High spenders

K-Means automatically creates these groups.

This process is called **Clustering**.

---

## What is Clustering?

Clustering means:

Grouping similar objects together.

Example:

Students with similar marks

Customers with similar spending

Movies with similar genres

---

## Definition of K-Means

K-Means is an unsupervised machine learning algorithm used to divide data into K clusters based on similarity.

Memorize this definition.

---

## What does K mean?

K = Number of clusters.

Example:

K = 2

Create 2 groups.

K = 3

Create 3 groups.

---

## Real Life Example

Suppose marks are:

10  
12  
15  
80  
85  
90

Choose:

$K = 2$

Output:

Cluster 1  
10 12 15

Cluster 2  
80 85 90

Simple.

---

## How K-Means Works

There are 4 important steps.

---

### Step 1: Choose K

Example:

$K = 2$

Need 2 clusters.

---

## Step 2: Select Centroids

Centroid = Center point of cluster.

Suppose:

Centroid 1 = 10

Centroid 2 = 90

---

## Step 3: Assign Data Points

Assign every point to nearest centroid.

Example:

10 → Cluster 1

12 → Cluster 1

15 → Cluster 1

80 → Cluster 2

85 → Cluster 2

90 → Cluster 2

---

## Step 4: Recalculate Centroids

Find new center.

Cluster 1:

$$(10+12+15)/3$$

$$= 12.3$$

Cluster 2:

$$(80+85+90)/3$$

$$= 85$$

New centroids:

12.3

85

---

Repeat until centroids stop changing.

Then algorithm ends.

---

## Flowchart

Choose K



Initialize Centroids



Assign Points



Calculate New Centroids



Centroids Changed?



Yes ↓

Repeat

No ↓

Stop

Draw this if asked algorithm.

---

## Easy Visualization

Imagine a classroom.

Teacher says:

Make 3 groups based on height.

Students automatically form:

Short

Medium

Tall

This is exactly clustering.

---

## Why K-Means is Called Unsupervised?

Supervised Learning:

Input + Correct Answer

Example:

Photo → Dog

Photo → Cat

Already labelled.

---

K-Means:

Only data given.

No labels.

Algorithm discovers groups itself.

Hence:

Unsupervised Learning

---

# Applications

## Customer Segmentation

Group customers.

---

## Recommendation Systems

Netflix

Amazon

---

## Image Segmentation

Grouping pixels.

---

## Medical Research

Grouping patients.

---

## Market Analysis

Finding customer categories.

---

# Advantages

## 1. Simple

Easy to understand.

## 2. Fast

Works efficiently.

## 3. Scalable

Can handle large datasets.

## 4. Easy Implementation

Widely used.

---

## Disadvantages

### 1. Need K in Advance

Must know number of clusters.

---

### 2. Sensitive to Initial Centroids

Different starting points can give different answers.

---

### 3. Poor with Outliers

Extreme values affect results.

---

### 4. Assumes Circular Clusters

Not ideal for irregular shapes.

---

## Difference Between Classification and Clustering

| Classification   | Clustering            |
|------------------|-----------------------|
| Supervised       | Unsupervised          |
| Labels available | No labels             |
| Predict class    | Discover groups       |
| Cat/Dog          | Group similar objects |

Very common theory question.

---

## 5 Marker Answer Structure

1. Definition
2. Diagram/Flowchart
3. Algorithm Steps
4. Applications
5. Advantages & Disadvantages

Done.

---

## Memory Trick

Think:

K = Number of Groups

Means = Center of Group

Hence:

K-Means

=

Make K Groups  
using Centers

---

## Ultra-Short Revision

Choose K

Choose Centroids

Assign Points

Update Centroids

Repeat Until Stable

This 5-step sequence alone can fetch most marks if you explain it properly.

Next: **Machine Learning** 🤖 (the parent topic from which supervised, unsupervised, and reinforcement learning come). This is another very likely exam question. 🔥

Sources

Next

## Topic 5: Machine Learning 🤖🧠

This is probably the most important concept in modern AI.

Everything you hear today:

- ChatGPT
- Netflix Recommendations
- YouTube Suggestions
- Self-Driving Cars
- Face Recognition

all use Machine Learning.

---

## First Understand the Problem

Traditional Programming:

Input + Rules

↓

Output

Example:

Marks > 40

Pass

Marks < 40

Fail

You manually write rules.

---

But what if there are millions of rules?

Example:

Identify Dog

Identify Cat

Identify Human Face

Identify Cancer

Impossible to manually write every rule.

So instead:

👉 We let the computer learn from data.

That's Machine Learning.

---

# Definition

Machine Learning is a branch of AI that enables computers to learn from data and improve their performance without being explicitly programmed.

Exam definition.

---

## Easy Example

Suppose you show a computer:

1000 Dog Photos

1000 Cat Photos

After learning patterns:

New Photo

↓

Computer says:

Dog

Nobody explicitly told every rule.

The machine learned.

---

## Real-Life Example

How does YouTube know what you want?

Because it learns:

What you watch

What you like

What you skip

How long you watch

Then predicts:

You may like this video

That's ML.

---

## Types of Machine Learning

Most Important Part.

There are 3 major types.

---

### 1. Supervised Learning

Learning using labelled data.

Example:

Image → Dog

Image → Cat

Image → Dog

Correct answers already available.

Machine learns relationship.

---

### Example

Predict:

Pass / Fail

Spam / Not Spam

Disease / No Disease

---

## 2. Unsupervised Learning

No labels provided.

Machine finds hidden patterns itself.

Example:

Customer Data

Machine discovers:

High Spenders

Medium Spenders

Low Spenders

without being told.

---

### Example

K-Means Clustering

This is why K-Means belongs here.

---

## 3. Reinforcement Learning

Learning through rewards and punishments.

Example:

Training a dog 🐶

Correct Action

Reward

Wrong Action

Penalty

Eventually:

Dog learns behavior

Same with AI.

## Example

# Chess AI

### Winning Move:

Reward +10

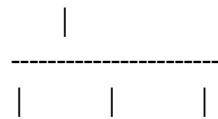
Bad Move:

Penalty -10

Eventually becomes stronger.

## Diagram

# Machine Learning



Supervised Unsupervised Reinforcement

Easy exam diagram.

# ML Process

## Data Collection



## Data Training



## Model Creation



## Testing



## Prediction

---

# Applications

## Netflix

Movie recommendations.

---

## Amazon

Product recommendations.

---

## Gmail

Spam detection.

---

## Healthcare

Disease prediction.

---

## Finance

Fraud detection.

---

## Self Driving Cars

Road understanding.

---

# Advantages

## 1. Learns Automatically

No need for explicit rules.

## 2. Handles Huge Data

Works on massive datasets.

## 3. Improves Over Time

More data → Better results.

## 4. Accurate Predictions

Can outperform humans in some tasks.

---

# Disadvantages

## 1. Requires Large Data

Need lots of examples.

---

## 2. Expensive Training

High computational cost.

---

## 3. Can Be Biased

Bad data → Bad predictions.

---

## 4. Difficult to Explain

Some models act like black boxes.

---

## Difference: AI vs ML

| AI          | ML           |
|-------------|--------------|
| Broad field | Subset of AI |

| AI                                | ML                            |
|-----------------------------------|-------------------------------|
| Makes machines intelligent        | Makes machines learn          |
| Includes search, logic, reasoning | Focuses on learning from data |

Memory trick:

AI  
└─ ML  
   └─ Deep Learning

---

## Difference: Supervised vs Unsupervised

| Supervised       | Unsupervised      |
|------------------|-------------------|
| Labels available | No labels         |
| Teacher present  | No teacher        |
| Predict output   | Discover patterns |
| Classification   | Clustering        |

## 5 Marker Answer Structure

1. Definition
  2. Types
    - o Supervised
    - o Unsupervised
    - o Reinforcement
  3. Applications
  4. Advantages
  5. Disadvantages
-

## Exam Hack

If they ask:

**"What is Machine Learning?"**

Always write:

Definition

Types of ML

Applications

Advantages

Even if not explicitly asked.

Profs love it.

---

## Ultra-Short Revision

Machine Learning

=

Learning from Data

Types:

1. Supervised
  2. Unsupervised
  3. Reinforcement
- 

Now only **Perception and Action** is left in Unit 4.

Good news: it's one of the shortest topics in the entire syllabus and can be finished in 10 minutes. 

## Topic 6: Perception and Action 🙄➡️⚙️

Congratulations 🎉

This is the **last topic of Unit 4** and honestly one of the easiest.

---

### First Understand the Idea

Imagine a human.

What do we do all day?

See things

Hear sounds

Feel surroundings

↓

Take decisions

↓

Perform actions

Example:

You see a speeding truck 🚚

Perception:

Truck is coming fast

Action:

Move away

Simple.

AI systems work exactly the same way.

---

# Definition

Perception is the process of acquiring information from the environment, while Action is the response performed by an intelligent agent based on that information.

Exam definition.

---

## Simple AI Example

Self-driving car.

### Perception

Car observes:

Road  
Traffic Light  
Pedestrians  
Other Vehicles

using:

Cameras  
Sensors  
LIDAR  
Radar

---

### Action

Car decides:

Turn Left

Brake

Accelerate

Stop

---

# The Basic Flow

Environment



Perception



Decision Making



Action



Environment

This is the most important diagram.

---

## Human Example

Imagine:

Room Temperature = 40°C

Perception

You feel hot.

Decision

Need cooling.

Action

Switch ON fan.

---

## AI Agent Example

Robot Vacuum Cleaner 

Perception

Sensors detect:

Dust  
Obstacle  
Wall

## Decision

Need cleaning

## Action

Move Forward  
Clean Floor  
Turn Right

---

# Components of Perception

AI perceives environment using sensors.

Examples:

## Camera

Sees objects.

---

## Microphone

Hears sounds.

---

## Temperature Sensor

Detects heat.

---

## GPS

Detects location.

---

# Components of Action

AI acts using actuators.

Examples:

## Wheels

Move robot.

---

## Robotic Arms

Pick objects.

---

## Speakers

Generate sound.

---

## Motors

Produce movement.

---

# Perception vs Action

| Perception          | Action             |
|---------------------|--------------------|
| Input               | Output             |
| Gather information  | Perform task       |
| Uses Sensors        | Uses Actuators     |
| Observe Environment | Change Environment |

# Real-Life Applications

## Self Driving Cars

Perceive road → Take driving actions.

---

## Robots

Perceive obstacles → Move safely.

---

## Smart Homes

Detect temperature → Adjust AC.

---

## Industrial Automation

Detect defects → Remove defective product.

---

# Advantages

## 1. Makes Agents Intelligent

Allows understanding of surroundings.

## 2. Enables Decision Making

Agent can react properly.

## 3. Supports Automation

Essential for robotics.

---

# Disadvantages

## 1. Sensor Errors

Wrong perception → Wrong action.

## 2. Expensive Hardware

Good sensors cost money.

## 3. Complex Environments

Difficult to interpret real-world situations.

---

# Easy Memory Trick

Think:

Eyes = Perception

Hands = Action

AI follows the same principle.

Sense

Think

Act

---

# 5 Marker Answer Structure

1. Definition
2. Perception Explanation
3. Action Explanation
4. Diagram
5. Applications
6. Advantages

Done.

---

## One-Line Revision ⚡

**Perception = Observe Environment**

**Action = Respond to Environment**

Perceive → Think → Act

---

## UNIT 4 COMPLETE

Final Revision Sheet

### Fuzzy Logic

- Handles uncertainty
- Fuzzification → Rules → Inference → Defuzzification

### Expert Systems

- Knowledge Base
- Inference Engine
- User Interface

### Bayesian Networks

- Nodes
- Edges
- Probabilities
- Reasoning under uncertainty

### K-Means

- Unsupervised Learning

- Choose K
- Assign Clusters
- Update Centroids

## Machine Learning

- Supervised
- Unsupervised
- Reinforcement

## Perception & Action

- Sensors → Perception
  - Decision → Action
  - Actuators
- 

## High Probability Ranking (based on 2024 + 2025 papers + notes)

### 🔥 S-Tier (Almost Certain)

1. Expert Systems
2. Bayesian Networks
3. K-Means
4. Machine Learning

### ★ A-Tier

5. Fuzzy Logic
6. Alpha-Beta Pruning (Unit 3)

### ✨ B-Tier

7. Perception & Action

Unit 4 is now exam-ready. 😎

My suggestion: jump to **Unit 3 next** because Alpha-Beta, Minimax, NLP, EBL and Neural vs Genetic Learning are heavily favored by past papers.

what is machine learning? Discuss the issues in machine learning and the steps required for selecting right machine learning algorithm and what are Bayesian networks in AI and how to solve them?

These are **very exam-oriented questions**, so I'll teach them in a way that helps you write a 5–7.5 marker answer.

---

## Q1. What is Machine Learning? Discuss the issues in Machine Learning and the steps required for selecting the right Machine Learning algorithm.

### Machine Learning Definition

Machine Learning (ML) is a branch of Artificial Intelligence that enables computers to learn from data and improve their performance without being explicitly programmed.

### Example

Suppose we show a computer:

1000 Dog Images

1000 Cat Images

The computer learns patterns and can classify a new image as a dog or cat.

---

## Issues in Machine Learning

Think of ML like teaching a student.

If teaching material is bad, student performs badly.

Similarly, ML faces several issues.

---

## 1. Poor Quality Data

Garbage In → Garbage Out

Example:

Wrong medical records

Incomplete customer data

ML model learns wrong patterns.

---

## 2. Insufficient Training Data

Less data means less learning.

Example:

Teaching a child only 2 math problems and expecting him to solve 100.

---

## 3. Overfitting

Model memorizes training data instead of learning.

Example:

Student memorizes answers but fails when question changes.

---

## 4. Underfitting

Model is too simple.

Fails to learn important patterns.

Example:

Studying only chapter names before exam.

---

## 5. High Computational Cost

Training large models requires:

Powerful CPUs

GPUs

Memory

Time

---

## 6. Feature Selection Problem

Not every input feature is useful.

Example:

For predicting house price:

Useful:

Area

Location

Bedrooms

Not Useful:

Owner's favorite color

---

## 7. Bias and Fairness

If training data is biased,

output becomes biased.

---

## Steps for Selecting the Right ML Algorithm

Very important theory question.

---

## Step 1: Understand the Problem

Ask:

Prediction?

Classification?

Clustering?

Example:

Spam detection → Classification

---

## Step 2: Analyze Data

Check:

Amount of Data

Quality of Data

Missing Values

---

## Step 3: Determine Learning Type

### Supervised

Labels available.

Example:

Dog/Cat

Pass/Fail

---

### Unsupervised

No labels.

Example:

Customer Segmentation

---

## Reinforcement

Reward/Penalty system.

Example:

Game Playing

---

## Step 4: Evaluate Algorithm Complexity

Choose algorithm according to:

Time

Memory

Resources

---

## Step 5: Train and Test

Train model.

Measure:

Accuracy

Precision

Recall

---

## Step 6: Compare Multiple Algorithms

Example:

Decision Tree

Neural Network

SVM

Choose best performing one.

---

## Exam Conclusion

Selection of the right ML algorithm depends on problem type, data quality, available resources, and desired accuracy.

---

## Easy Memory Trick

Issues:

Data

Overfitting

Underfitting

Cost

Bias

Features

---

## Q2. What are Bayesian Networks in AI and How to Solve Them?

---

### Definition

A Bayesian Network is a probabilistic graphical model that represents variables and their dependencies using a Directed Acyclic Graph (DAG).

---

### Basic Structure

Example:

Cloudy

↓

Rain

↓

Wet Grass

Interpretation:

Cloudy affects Rain

Rain affects Wet Grass

---

## Components

### 1. Nodes

Represent variables.

Example:

Cloudy

Rain

Wet Grass

---

### 2. Directed Edges

Represent dependency.

Cloudy  $\rightarrow$  Rain

Rain  $\rightarrow$  Wet Grass

---

### 3. Probability Table

Stores probabilities.

Example:

$P(\text{Rain} \mid \text{Cloudy}) = 0.8$

Meaning:

If cloudy,

80% chance of rain.

---

# How to Solve Bayesian Network Questions

Most exam questions are simple probability calculations.

---

## Formula

Bayes Theorem:

---

## Example

Given:

$$P(\text{Disease}) = 0.01$$

$$P(\text{Test Positive} \mid \text{Disease}) = 0.99$$

$$P(\text{Test Positive}) = 0.05$$

Find:

$$P(\text{Disease} \mid \text{Test Positive})$$

Using Bayes theorem:

$$= (0.99 \times 0.01) / 0.05$$

$$= 0.0099 / 0.05$$

$$= 0.198$$

Answer:

$$19.8\%$$

---

## Joint Probability Method

For a chain:

$$A \rightarrow B \rightarrow C$$

Joint Probability:

$$P(A, B, C) = P(A)P(B|A)P(C|A, B)$$

Example:

$$P(\text{Cloudy}) = 0.5$$

$$P(\text{Rain} | \text{Cloudy}) = 0.8$$

$$P(\text{WetGrass} | \text{Rain}) = 0.9$$

Then:

$$P(\text{Cloudy}, \text{Rain}, \text{WetGrass})$$

$$= 0.5 \times 0.8 \times 0.9$$

$$= 0.36$$

---

## Applications

### Medical Diagnosis

Disease prediction.

### Weather Forecasting

Rain prediction.

### Spam Detection

Email classification.

### Risk Analysis

Financial prediction.

---

## 7.5 Marker Answer Structure

### Bayesian Networks

1. Definition
  2. DAG Diagram
  3. Components
    - o Nodes
    - o Edges
    - o Probability Tables
  4. Bayes Theorem
  5. Example Calculation
  6. Applications
  7. Advantages
- 

## Super Short Exam Revision ⚡

### Machine Learning

Learn from Data

Issues:

- Poor Data
- Overfitting
- Underfitting
- Cost
- Bias

Selection:

Problem → Data → Learning Type → Train → Evaluate

### Bayesian Network

Nodes = Variables

Edges = Dependencies

Probabilities = Uncertainty

Formula:

$$P(A | B) = P(B | A)P(A)/P(B)$$

These two questions together are easily worth 10–15 marks in an end-sem paper. 🎯

Lets start with U 2 now

Lesssgooo 🔥

## Unit 2 Strategy

Unit 2 is the **highest theory-weightage unit**.

The examiner usually asks direct questions from headings.

Order we'll study:

1. Knowledge Representation
2. Knowledge Representation Issues + Mapping + Frame Problem
3. Predicate Logic
4. Facts in Logic + ISA & Instance Relationship
5. Resolution ★★★★★
6. Procedural vs Declarative Knowledge
7. Matching
8. Control Knowledge
9. Symbolic Reasoning Under Uncertainty
10. Non-Monotonic Reasoning ★★★★★

# Topic 1: Knowledge Representation (KR)

Before AI can think...

it must **store knowledge**.

Just like humans.

Example:

You know:

Delhi is capital of India.

Dog is an animal.

Birds can fly.

Your brain stores this knowledge.

Similarly AI needs a way to store knowledge.

That process is called:

## Knowledge Representation

---

### Definition

Knowledge Representation is the method of organizing and storing knowledge in a form that a computer can understand and use for reasoning.

Exam definition.

---

# Why Do We Need Knowledge Representation?

Imagine an AI doctor.

Without knowledge:

Patient has fever

AI can't do anything.

But if AI knows:

IF Fever AND Cough

THEN Flu

Now it can reason.

Hence:

Knowledge

+

Reasoning

=

Intelligence

---

## Components of Knowledge Representation

Knowledge Representation has two jobs:

### 1. Store Knowledge

Example:

Dog is an Animal

---

### 2. Use Knowledge

Example:

Dog is an Animal

Animal is Living Thing

AI concludes:

Dog is Living Thing

This process is reasoning.

---

## Types of Knowledge

Very Important ★★ ★

---

### 1. Declarative Knowledge

"Knowing What"

Facts.

Example:

Delhi is capital of India

$2 + 2 = 4$

---

### 2. Procedural Knowledge

"Knowing How"

Steps.

Example:

How to ride a bicycle

How to solve a puzzle

---

### 3. Heuristic Knowledge

Rules of thumb.

Example:

Choose shortest route first

Not always correct.

Usually correct.

---

## 4. Meta Knowledge

Knowledge about knowledge.

Example:

I know that AI is a subject.

---

# Techniques of Knowledge Representation

Most asked exam question.

There are 4 techniques.

---

## 1. Logical Representation

Uses logic.

Example:

Human(Sam)

$\text{Human}(x) \rightarrow \text{Mortal}(x)$

Therefore:

Mortal(Sam)

---

## 2. Semantic Network

Stores knowledge as graph.

Example:

Dog → Animal

Animal → Living Thing

Easy visualization.

---

### 3. Frame Representation

Stores knowledge like tables.

Example:

Dog

Color = Brown

Legs = 4

Sound = Bark

Similar to database records.

---

### 4. Production Rules

Uses IF-THEN rules.

Example:

IF Fever

THEN Disease

Used in Expert Systems.

---

## Diagram

Knowledge



Knowledge Representation



Reasoning



Decision Making

Simple but effective.

---

## Real-Life Example

Google Maps.

Stores:

Roads

Traffic

Locations

Then reasons:

Shortest Route

Without KR, impossible.

---

## Advantages

### 1. Organizes Knowledge

Easy storage.

---

### 2. Supports Reasoning

AI can think.

---

### 3. Improves Decision Making

Better conclusions.

---

#### 4. Reusability

Knowledge can be reused.

---

## Disadvantages

#### 1. Difficult for Complex Knowledge

Real world is huge.

---

#### 2. Requires Maintenance

Knowledge must be updated.

---

#### 3. Ambiguity

Human knowledge is often vague.

---

## Memory Trick

Think:

Human Brain  
Stores Knowledge

AI Brain  
Uses Knowledge Representation

---

## 5 Marker Answer Structure

#### 1. Definition

2. Need for KR
3. Types of Knowledge
4. Techniques of KR
5. Advantages

Done.

---

## Ultra-Short Revision ⚡

Knowledge Representation

= Storing knowledge in a form  
computers can understand

Techniques:

1. Logic
  2. Semantic Network
  3. Frames
  4. Production Rules
- 

🔥 Next Topic: **Knowledge Representation Issues + Mapping + Frame Problem**

This is one of the most boring-looking topics, but examiners love asking it because students usually skip it.

Next

## Topic 2: Knowledge Representation Issues, Mapping & Frame Problem 🧩

This topic looks scary because of the names.

But honestly, it's mostly common sense.

---

## Part 1: Knowledge Representation Issues

### First Understand the Problem

Suppose I tell AI:

A dog is an animal.

Easy.

Now suppose I tell AI:

A dog is loyal.

Dogs bark.

Some dogs are pets.

Some dogs are police dogs.

Now knowledge becomes complex.

Question:

👉 How should AI store all this knowledge correctly?

These difficulties are called **Knowledge Representation Issues**.

---

## Definition

Knowledge Representation Issues are challenges involved in storing, organizing, and using knowledge efficiently in an AI system.

---

## Main Issues

### 1. What to Represent?

AI must decide what knowledge is important.

Example:

For a self-driving car:

Important:

Road

Traffic

Pedestrian

Not Important:

Color of nearby bird

---

## 2. How to Represent Knowledge?

Should AI use:

Logic?

Frames?

Semantic Networks?

Rules?

Choosing representation is a challenge.

---

## 3. Incomplete Knowledge

AI rarely knows everything.

Example:

Patient has fever.

AI still doesn't know disease.

Need reasoning.

---

## 4. Uncertain Knowledge

Real world isn't always certain.

Example:

Clouds

doesn't always mean:

Rain

Need probabilistic reasoning.

---

## 5. Updating Knowledge

Knowledge changes.

Example:

Pluto = Planet

Old knowledge.

Now:

Pluto  $\neq$  Planet

AI must update knowledge.

---

## Memory Trick

KR Issues:

What?

How?

Incomplete?

Uncertain?

Updating?

---

## Part 2: Mapping

This topic sounds fancy but is very easy.

---

## What is Mapping?

Mapping means:

Connecting real-world objects to their representation inside AI.

---

### Example

Real World:

Dog

Inside AI:

Animal(Dog)

AI converts real-world objects into symbols.

This conversion is called Mapping.

---

### Another Example

Real World:

Car has 4 wheels.

AI Representation:

Car(Wheels = 4)

---

## Why Mapping is Important?

Without mapping:

Human understands object

Computer understands nothing

Mapping acts as translator.

---

# Easy Exam Definition

Mapping is the process of converting real-world knowledge into a formal representation understandable by a computer.

---

## Part 3: Frame Problem

This is the most important topic from this chapter.

Very commonly asked.

---

## First Understand the Problem

Imagine a robot.

Initial State:

Book on Table

Robot performs action:

Pick Book

Now:

Book is in Hand

Question:

What happened to everything else?

Did:

Table change?

Room change?

Chair change?

Fan change?

No.

Only the book's position changed.

---

## The Problem

When an action occurs,

AI must determine:

What changed?

and

What remained unchanged?

This difficulty is called the Frame Problem.

---

## Definition

The Frame Problem is the problem of representing the effects of actions without explicitly describing every unchanged fact.

Memorize this definition.

---

## Example

Before:

Light = OFF

Door = Closed

Fan = OFF

Action:

Open Door

After:

Door = Open

Everything else remains same.

AI should know:

Light still OFF

Fan still OFF

without being told again.

This is Frame Problem.

---

## Why is it Difficult?

Real world contains millions of facts.

Imagine telling AI:

Door changed

Fan unchanged

Light unchanged

Chair unchanged

Wall unchanged

Window unchanged

...

Impossible.

---

## Real-Life Example

Self-driving car.

Action:

Turn Left

What changes?

Direction

What stays same?

Engine

Color

Passengers

Fuel

AI must identify this automatically.

---

## Solutions (Just Know Names)

For exam:

### 1. Frame Axioms

Explicitly mention unchanged facts.

---

### 2. Default Reasoning

Assume things stay same unless specified.

Most common answer.

---

### 3. Non-Monotonic Reasoning

Allow knowledge to change when new facts arrive.

(You'll study this later in Unit 2.)

---

## 5 Marker Answer Structure

If asked:

**"Explain Frame Problem."**

Write:

1. Definition
  2. Example (Book/Table or Door)
  3. Why it is difficult
  4. Solutions
  5. Conclusion
- 

## Quick Revision

### Knowledge Representation Issues

What to represent

How to represent

Incomplete knowledge

Uncertain knowledge

Updating knowledge

### Mapping

Real World

↓

AI Symbols

### Frame Problem

Action happens

AI must know:

What changed?

What stayed same?

---

## Exam Prediction

Among these three:

Knowledge Representation Issues  
Mapping  
Frame Problem

👉 **Frame Problem** has the highest probability of appearing as a direct theory question.

---

Next: **Predicate Logic** 🔥

This is where Unit 2 becomes scoring because once you understand Predicate Logic, you'll automatically understand:

- Facts in Logic
- ISA Relationship
- Instance Relationship
- Resolution

which together form nearly half of Unit 2. 🚀

Lessgoo

## Topic 3: Predicate Logic 🧠⚖️

This is the **foundation of the rest of Unit 2**.

If you understand Predicate Logic today, then:

- Facts in Logic ✅
- ISA Relationship ✅
- Instance Relationship ✅
- Resolution ✅

become much easier.

---

## First Understand the Problem

Suppose I say:

All humans are mortal.

Ram is a human.

Can AI conclude:

Ram is mortal.

YES.

But how?

AI needs a formal language.

That language is called **Predicate Logic**.

---

## Why Not Simple English?

English is ambiguous.

Example:

I saw a man with a telescope.

Who has telescope?

Me?

Man?

Confusing.

Computers hate ambiguity.

So AI uses Predicate Logic.

---

# Definition

Predicate Logic is a formal system used to represent facts, objects, properties, and relationships and to perform logical reasoning.

Exam definition.

---

## Components of Predicate Logic

There are 4 things.

---

### 1. Predicate

A property or relationship.

Example:

Human(Ram)

Meaning:

Ram is a Human

Here:

Human = Predicate

Ram = Object

---

More examples:

Student(Anmol)

Teacher(Rahul)

Likes(Alice,Bob)

Meaning:

Alice likes Bob

---

## 2. Constant

Specific object.

Examples:

Ram

Delhi

India

---

## 3. Variable

Represents any object.

Examples:

x

y

z

---

Example:

Human(x)

means:

x can be any human

---

## 4. Function

Maps one object to another.

Example:

Father(Ram)

Output:

Shyam

Meaning:

Father of Ram is Shyam

---

## Quantifiers

Most important part.

Two quantifiers only.

---

### Universal Quantifier ( $\forall$ )

Means:

FOR ALL

Symbol:

$\forall$

---

Example:

All humans are mortal.

Predicate Logic:

$\forall x [\text{Human}(x) \rightarrow \text{Mortal}(x)]$

Read:

For every x,

if x is human,

then x is mortal.

---

## Existential Quantifier ( $\exists$ )

Means:

THERE EXISTS

Symbol:

$\exists$

---

Example:

Some student is intelligent.

Predicate Logic:

$\exists x [\text{Student}(x) \wedge \text{Intelligent}(x)]$

Meaning:

There exists a student  
who is intelligent.

---

## Important Symbols

| Symbol        | Meaning      |
|---------------|--------------|
| $\wedge$      | AND          |
| $\vee$        | OR           |
| $\neg$        | NOT          |
| $\rightarrow$ | IMPLIES      |
| $\forall$     | FOR ALL      |
| $\exists$     | THERE EXISTS |

Memorize this table.

---

## Example 1 ★

Statement:

Ram is a student.

Predicate Form:

$\text{Student}(\text{Ram})$

---

## Example 2 ★

Statement:

All birds fly.

Predicate Form:

$\forall x [\text{Bird}(x) \rightarrow \text{Fly}(x)]$

---

## Example 3 ★

Statement:

Some birds cannot fly.

Predicate Form:

$\exists x [\text{Bird}(x) \wedge \neg \text{Fly}(x)]$

---

## Why Predicate Logic is Better than Propositional Logic?

Suppose:

Ram is human

Shyam is human

Aman is human

Propositional Logic:

P

Q

R

No relationship.

---

Predicate Logic:

Human(Ram)

Human(Shyam)

Human(Aman)

Much richer information.

---

## Predicate Logic vs Propositional Logic

| Predicate Logic          | Propositional Logic   |
|--------------------------|-----------------------|
| More powerful            | Less powerful         |
| Uses variables           | No variables          |
| Represents relationships | Represents facts only |
| Supports quantifiers     | No quantifiers        |

Very common theory question.

---

# Topic 4: Facts in Logic, ISA Relationship & Instance Relationship 🔥

This topic is directly built on Predicate Logic.

Once you understand these 3 concepts, you're ready for **Resolution** (the most important Unit 2 topic).

---

## Part 1: Facts in Logic

### What is a Fact?

A fact is a statement that is true.

Examples:

Delhi is capital of India.

Ram is a student.

Dog is an animal.

AI stores these facts in Predicate Logic.

---

### Examples

Statement:

Ram is a student.

Predicate Form:

Student(Ram)

---

Statement:

Delhi is a city.

Predicate Form:

City(Delhi)

---

Statement:

Dog is an animal.

Predicate Form:

Animal(Dog)

These are called **facts in logic**.

---

## Why Facts Matter?

AI reasons using facts.

Example:

Facts:

Human(Ram)

$\forall x [\text{Human}(x) \rightarrow \text{Mortal}(x)]$

AI concludes:

Mortal(Ram)

Without facts, reasoning is impossible.

---

## Part 2: ISA Relationship

Very frequently asked.

---

### What is ISA?

ISA means:

IS A TYPE OF

or

BELONGS TO A CLASS

---

## Example

Dog IS A Animal

Representation:

Dog ISA Animal

---

Another example:

Car ISA Vehicle

---

Another example:

Student ISA Human

---

## Visualization

Animal

↑

Dog

Meaning:

Dog is a type of Animal.

---

## Important Property

Inheritance.

If:

Animal can breathe

and

Dog ISA Animal

Then automatically:

Dog can breathe

This is called inheritance.

---

## Example

Bird ISA Animal

Animal ISA LivingThing

AI concludes:

Bird ISA LivingThing

---

## Part 3: Instance Relationship

Students confuse this with ISA.

Examiner loves asking the difference.

---

## What is Instance?

Instance means:

Specific example of a class.

---

Example:

Dog = Class

Tommy = Specific Dog

Therefore:

Tommy INSTANCE-OF Dog

---

Another example:

Student = Class

Anmol = Instance

Representation:

Anmol INSTANCE-OF Student

---

## Easy Visualization

Animal

↑

Dog

↑

Tommy

Here:

Dog ISA Animal

Tommy INSTANCE-OF Dog

---

## Difference Between ISA and Instance

Most important table.

| ISA                          | Instance                        |
|------------------------------|---------------------------------|
| Relationship between classes | Relationship between object and |

| ISA            | Instance              |
|----------------|-----------------------|
|                | class                 |
| Dog ISA Animal | Tommy INSTANCE-OF Dog |
| Class → Class  | Object → Class        |
| Generalization | Specific Example      |

## Exam Example

Given:

Animal can breathe

Dog ISA Animal

Tommy INSTANCE-OF Dog

What can AI conclude?

Step 1:

Dog can breathe

(using ISA)

Step 2:

Tommy can breathe

(using Instance)

---

## Knowledge Hierarchy Diagram ★★

Draw this.

Living Thing

↑

Animal

↑

Dog

↑  
Tommy

Relations:

Animal ISA Living Thing

Dog ISA Animal

Tommy INSTANCE-OF Dog

Easy marks.

---

## Real-Life Example

Vehicle

↑  
Car

↑  
Honda City

Here:

Car ISA Vehicle

Honda City INSTANCE-OF Car

---

## 5 Marker Answer Structure

Facts in Logic

- Definition
- Examples

ISA Relationship

- Definition
- Example

- Inheritance

## Instance Relationship

- Definition
  - Example
  - Difference from ISA
- 

## Quick Revision ⚡

### Fact

Student(Ram)

True statement.

---

### ISA

Dog ISA Animal

Class → Class

---

### Instance

Tommy INSTANCE-OF Dog

Object → Class

---

## Memory Hack 🧠

Think:

Animal

↑

Dog

↑  
Tommy

Dog ISA Animal

Tommy INSTANCE-OF Dog

If you remember this one diagram, you'll never confuse ISA and Instance again.

---

🔥 Next Topic: **Resolution**

This is arguably the **most important and most examinable topic in Unit 2**.

After Resolution, almost 60-70% of Unit 2's scoring content is covered. 🚀

nexxt

## Topic 5: Resolution 🔥 🔥 🔥

Welcome to the king of Unit 2 👑

If I had to bet on one topic from Unit 2 appearing in the paper, it'd be **Resolution**.

Appeared directly in your notes and is a classic AI endsem favorite.

AI Endsem Notes (Complete) (1)

---

## First Understand Why We Need Resolution

Suppose AI knows:

All humans are mortal.

Ram is a human.

Can AI prove:

Ram is mortal.

Yes.

But AI needs a systematic method.

That method is called:

## Resolution

---

### Definition

Resolution is an inference rule used in Predicate Logic to derive conclusions by eliminating contradictory literals.

Exam definition.

---

### Human Version

Think like a detective 🕵️

You want to prove:

Ram is Mortal

Instead of proving it directly,

you assume:

Ram is NOT Mortal

and check if it creates a contradiction.

If contradiction occurs,

the original statement must be true.

---

### Resolution Works by Contradiction

Very important.

Steps:

1. Take given facts
  2. Negate the goal
  3. Apply resolution
  4. Get contradiction
  5. Goal proved
- Memorize this.
- 

## Example 1 (Most Important)

Given:

All humans are mortal.

Ram is human.

Prove:

Ram is mortal.

---

## Step 1: Convert to Predicate Logic

Statement 1:

$\forall x [\text{Human}(x) \rightarrow \text{Mortal}(x)]$

Statement 2:

$\text{Human}(\text{Ram})$

Goal:

$\text{Mortal}(\text{Ram})$

---

## Step 2: Convert Implication

Remember:

$$\begin{aligned} A \rightarrow B \\ = \\ \neg A \vee B \end{aligned}$$

Therefore:

$$\neg \text{Human}(x) \vee \text{Mortal}(x)$$

---

## Step 3: Negate Goal

Goal:

$\text{Mortal}(\text{Ram})$

Negation:

$$\neg \text{Mortal}(\text{Ram})$$

---

## Step 4: Clauses

We now have:

C1:

$$\neg \text{Human}(x) \vee \text{Mortal}(x)$$

C2:

$\text{Human}(\text{Ram})$

C3:

$$\neg \text{Mortal}(\text{Ram})$$

---

## Step 5: Resolution

Resolve:

C1 and C2

Human cancels.

Result:

Mortal(Ram)

---

Now resolve:

Mortal(Ram)

and

$\neg$ Mortal(Ram)

Result:

☐

(empty clause)

---

## Empty Clause Means

Contradiction Found

Therefore:

Mortal(Ram) is TRUE

Goal proved.

---

## Visual Flow

Given Facts



Convert to Clauses



Negate Goal



Apply Resolution



Empty Clause

↓

Goal Proved

Draw this.

---

## Another Small Example

Given:

Dog(Tommy)

$\forall x [\text{Dog}(x) \rightarrow \text{Animal}(x)]$

Prove:

Animal(Tommy)

---

Convert:

$\neg \text{Dog}(x) \vee \text{Animal}(x)$

Dog(Tommy)

$\neg \text{Animal}(\text{Tommy})$

Resolve:

Dog(Tommy)

with

$\neg \text{Dog}(x) \vee \text{Animal}(x)$

gives

Animal(Tommy)

Resolve with:

$\neg \text{Animal}(\text{Tommy})$

gives

□

Proved.

---

## Why Resolution is Powerful?

Instead of creating hundreds of rules,

AI uses one inference mechanism.

That's why expert systems and theorem provers use it.

---

## Advantages

### 1. Sound

Produces correct conclusions.

---

### 2. Complete

Can prove any logically valid statement.

---

### 3. Simple Mechanism

Single inference rule.

---

### 4. Widely Used

Used in automated reasoning.

---

# Disadvantages

## 1. Can Become Slow

Large knowledge bases.

---

## 2. Requires CNF Form

Statements must be converted first.

---

## 3. Computationally Expensive

Complex problems take time.

---

# Exam Question Pattern

Usually:

## Theory

What is Resolution?

Explain with example.

OR

## Numerical Style

Use Resolution to prove:

Socrates is Mortal

Ram is Mortal

etc.

---

# Super Important Formula

Implication Removal:

$$A \rightarrow B \equiv \neg A \vee B$$

This is used in almost every resolution problem.

---

## 7 Marker Answer Structure

1. Definition
  2. Principle of contradiction
  3. Resolution steps
  4. Worked example
  5. Empty clause explanation
  6. Advantages
- 

## Quick Revision

Resolution

1. Convert to clauses
  2. Negate goal
  3. Resolve clauses
  4. Get empty clause
  5. Goal proved
- 

## Memory Trick

Think:

Want to prove X

Assume NOT X

If contradiction occurs

Then X is true

That's literally the entire idea behind Resolution.

---

### 🔥 Next: **Procedural Knowledge vs Declarative Knowledge**

Very easy topic, very frequently asked as a 3–5 marker, and takes only 5 minutes to master. 😎

Sources

yes less gooo

## Topic 6: Procedural Knowledge vs Declarative Knowledge 🧠🔪

This is one of the easiest and most repeated theory questions in Unit 2.

Examiner loves asking:

Differentiate between Procedural Knowledge and Declarative Knowledge.

Easy 5 marks if you know the table.

---

## First Understand with a Real Example

Imagine I ask you:

### Question 1

What is the capital of India?

You answer:

Delhi

This is knowledge of a fact.

This is:

## Declarative Knowledge

---

### Question 2

How do you ride a bicycle?

Now you explain:

Balance

Pedal

Steer

Brake

This is knowledge of a process.

This is:

## Procedural Knowledge

---

# Declarative Knowledge

## Definition

Declarative Knowledge refers to knowledge about facts, objects, events, and concepts.

Simply:

Knowing WHAT

---

## Examples

Delhi is capital of India.

Dog is an animal.

$2 + 2 = 4$ .

Ram is a student.

All are facts.

---

## AI Representation

Student(Ram)

Capital(Delhi, India)

These are stored as facts.

---

# Procedural Knowledge

## Definition

Procedural Knowledge refers to knowledge about how to perform a task or procedure.

Simply:

Knowing HOW

---

## Examples

How to solve a puzzle

How to drive a car

How to sort numbers

How to bake a cake

These are procedures.

---

## AI Representation

Usually represented as:

Algorithms

Procedures

Rules

Instructions

---

## Easy Analogy

Imagine preparing Maggi.

---

## Declarative Knowledge

Maggi contains noodles.

Cooking time = 2 minutes.

Packet weight = 70g.

Facts.

---

## Procedural Knowledge

Boil water

Add noodles

Add masala

Cook 2 minutes

Steps.

---

## Comparison Table

This table is exam gold.

| Declarative Knowledge              | Procedural Knowledge        |
|------------------------------------|-----------------------------|
| Knowing WHAT                       | Knowing HOW                 |
| Stores facts                       | Stores procedures           |
| Easier to modify                   | Harder to modify            |
| Describes knowledge                | Describes actions           |
| Example: Delhi is capital of India | Example: How to reach Delhi |

Memorize this.

---

## AI Example

Suppose AI Doctor.

---

## Declarative

Fever is symptom of Flu.

Flu is caused by virus.

Facts.

---

## Procedural

Step 1: Check fever

Step 2: Check cough

Step 3: Diagnose disease

Procedure.

---

## Advantages of Declarative Knowledge

### 1. Easy to Understand

Human readable.

---

### 2. Easy to Update

Facts can be changed easily.

---

### 3. Supports Reasoning

Works well with logic systems.

---

# Advantages of Procedural Knowledge

## 1. Efficient Execution

Directly tells system what to do.

---

## 2. Useful for Problem Solving

Provides step-by-step solutions.

---

## 3. Faster Decision Making

No need to derive procedures.

---

# Disadvantages

## Declarative

May require separate reasoning mechanism.

---

## Procedural

Difficult to modify and maintain.

---

# Exam Question

## Differentiate Between Procedural and Declarative Knowledge

Write:

| Declarative | Procedural |
|-------------|------------|
| What        | How        |
| Facts       | Procedures |

| Declarative    | Procedural     |
|----------------|----------------|
| Easy to modify | Hard to modify |
| Logic based    | Action based   |

Add one example.

Done.

5/5 marks. 😎

---

## Memory Trick

Think:

### Declarative

I know that...

Examples:

Delhi is capital of India.

---

### Procedural

I know how...

Examples:

How to drive.

How to cook.

---

## Ultra Fast Revision

Declarative = WHAT

Procedural = HOW

That's literally the whole topic.

---

🔥 Next up:

## Matching & Control Knowledge

These are short topics, usually combined into one question, and we can finish both in 10–15 minutes before moving to the heavy hitters:

- Symbolic Reasoning Under Uncertainty
- Non-Monotonic Reasoning ★★★
- Statistical Reasoning

Lesssgooo 🚀🔥

lessfuckkingg go

## Topic 7: Matching & Control Knowledge 🔥

These are small topics, but examiners LOVE sneaking 2–3 markers from them.

---

### Part 1: Matching 🎯

#### First Understand the Idea

Suppose an Expert System has these rules:

Rule 1:

IF Fever AND Cough  
THEN Flu

Rule 2:

IF Chest Pain  
THEN Heart Disease

User enters:

Fever  
Cough

Question:

👉 How does AI know which rule to use?

It compares the given facts with stored rules.

This process is called:

## Matching

---

### Definition

Matching is the process of comparing current facts with stored rules in order to find applicable rules.

---

### Example

Knowledge Base:

IF Student studies  
THEN Pass

IF Student sleeps  
THEN Fail

Fact:

Student studies

Matching finds:

IF Student studies  
THEN Pass

and activates it.

---

## Simple Diagram

Facts



Matching



Applicable Rule Found



Inference

---

## Why Matching is Important?

Without matching:

AI has rules

AI has facts

But AI cannot connect them.

Matching is the bridge.

---

## Real-Life Example

Google Maps:

Traffic High

Road Closed

System matches conditions and chooses alternate route.

---

## Part 2: Control Knowledge

This sounds complicated but is actually very easy.

---

# First Understand

Suppose AI has:

1000 rules

Many rules match at the same time.

Example:

Rule 1 matches

Rule 2 matches

Rule 3 matches

Question:

👉 Which one should execute first?

Control Knowledge decides that.

---

## Definition

Control Knowledge is knowledge that guides the reasoning process by deciding which rule or action should be applied next.

---

## Easy Example

Suppose:

Rule 1:

Treat Fever

Rule 2:

Treat Heart Attack

Both match.

Common sense says:

Heart Attack first

Control Knowledge makes this decision.

---

## Real-Life Analogy

Imagine:

You have:

- Assignment
- Exam tomorrow
- Netflix

All are possible actions.

Your brain decides:

Exam first

That's Control Knowledge 🕶️

---

## Why is Control Knowledge Needed?

Without it:

AI gets confused.

Too many rules.

Too many possibilities.

Control Knowledge helps prioritize.

---

## Types of Control Strategies

Just know the names.

## 1. Priority Based

Important rules first.

---

## 2. Specificity Based

More specific rules first.

---

## 3. Recency Based

Newest information first.

---

# Difference Between Matching and Control Knowledge

| Matching               | Control Knowledge             |
|------------------------|-------------------------------|
| Finds applicable rules | Chooses which rule to execute |
| Fact ↔ Rule comparison | Rule selection                |
| First step             | Next step                     |
| "Which rules match?"   | "Which rule should run?"      |

★★★★ Very important table.

---

## Complete Flow

Facts  
↓  
Matching  
↓  
Many Rules Match  
↓  
Control Knowledge  
↓  
Choose Best Rule

↓  
Inference

Memorize this flow.

---

## Exam Answer Structure

### Matching

- Definition
- Example
- Importance

### Control Knowledge

- Definition
- Example
- Difference from Matching

Done.

---

## Quick Revision ⚡

### Matching

Facts + Rules

↓  
Find Matching Rule

---

### Control Knowledge

Many Rules Match

↓  
Choose Best Rule

---

## Memory Hack

Think of a dating app 😊

### Matching

Find all compatible people

### Control Knowledge

Choose who to text first

Exactly the same AI logic.

---

🔥🔥🔥 Now we enter the BIG BOY topics of Unit 2:

## Next: Symbolic Reasoning Under Uncertainty

After that:

- Non-Monotonic Reasoning ★★★★★
- Statistical Reasoning

These three are among the highest-probability theory questions remaining in Unit 2.

LESSSSSFUCKINGGOOOOO 🚀😞📖

go

## Topic 8: Symbolic Reasoning Under Uncertainty

This topic sounds intimidating.

But the idea is super simple.

---

# First Understand the Problem

Traditional Logic says:

True

OR

False

Example:

It is raining.

Either:

True

or

False

---

But real life isn't like that.

Suppose:

Dark Clouds

High Humidity

Strong Wind

Will it rain?



You can't say 100%.

You can only say:

Probably

Maybe

Likely

This is called:

# Reasoning Under Uncertainty

---

## Definition

Symbolic Reasoning Under Uncertainty is the process of making decisions or drawing conclusions when available information is incomplete, uncertain, or ambiguous.

---

## Why Does Uncertainty Exist?

AI rarely gets perfect information.

---

## Example 1: Medical Diagnosis

Patient has:

Fever

Headache

Could be:

Flu

COVID

Malaria

Dengue

AI isn't certain.

---

## Example 2: Weather Forecast

Clouds Present

Doesn't guarantee rain.

Only increases probability.

---

## Example 3: Self Driving Car

Sensor says:

Object Ahead

Could be:

Person

Bike

Pole

Need reasoning under uncertainty.

---

## Symbolic Reasoning

First understand "Symbolic".

AI stores knowledge as symbols.

Example:

Rain

Cloud

Disease

Human

Instead of raw numbers.

Then AI reasons using these symbols.

---

## Problem

Normal logic works only when information is certain.

Example:

Human(Ram)

All Humans are Mortal

Conclusion:

Mortal(Ram)

100% sure.

---

Real world isn't like that.

Hence uncertainty methods are needed.

---

## Methods Used

Very important.

---

### 1. Probability Theory

Uses likelihood.

Example:

$P(\text{Rain}) = 0.8$

Meaning:

80% chance of rain.

---

### 2. Bayesian Networks

You already studied this.

Example:

Cloudy

↓

Rain

Used when events influence each other.

---

### 3. Fuzzy Logic

Instead of:

Hot = Yes

Hot = No

Uses:

Hot = 0.7

Handles vagueness.

---

### 4. Certainty Factors

Used in Expert Systems.

Example:

Disease Confidence = 80%

instead of

Disease = TRUE

---

## Real Life Example

Suppose AI doctor knows:

If Fever  $\rightarrow$  Flu

Patient has:

Mild Fever

AI may say:

70% chance Flu

instead of:

Definitely Flu

This is uncertainty reasoning.

---

## Advantages

### 1. Works in Real World

Real world is uncertain.

---

### 2. Better Decisions

Handles incomplete information.

---

### 3. More Human-Like

Humans also reason probabilistically.

---

## Disadvantages

### 1. Complex Calculations

Need probability models.

---

### 2. Requires More Data

Need evidence and statistics.

---

### 3. Results Not Guaranteed

Predictions are probabilistic.

---

## Difference

| Normal Reasoning | Uncertain Reasoning |
|------------------|---------------------|
| True / False     | Probability         |
| Certain facts    | Incomplete facts    |
| Exact conclusion | Likely conclusion   |

## Exam Example

Question:

Patient has fever.

Normal Logic:

Fever  $\rightarrow$  Disease

Conclusion:

Disease

---

Uncertain Reasoning:

Fever  $\rightarrow$  60% Flu

Fever  $\rightarrow$  20% Dengue

Fever  $\rightarrow$  20% Viral

Much more realistic.

---

## 5 Marker Answer Structure

1. Definition
2. Need for uncertainty reasoning
3. Causes of uncertainty

#### 4. Methods

- o Probability
- o Bayesian Networks
- o Fuzzy Logic
- o Certainty Factors

#### 5. Applications

---

## Quick Revision

Real world  $\neq$  Certain

AI needs:  
Probability  
Bayesian Networks  
Fuzzy Logic  
Certainty Factors

---

## Memory Hack

Think:

Normal Logic:  
YES / NO

Uncertainty Logic:  
MAYBE 

---

 NEXT UP:

## Non-Monotonic Reasoning

This is the **single most important remaining topic in Unit 2.**

It gets asked directly, indirectly, and in short notes.

Once we finish that, only Statistical Reasoning remains and Unit 2 is DONE. 🚀



next

## Topic 9: Non-Monotonic Reasoning 🔥👑

This is probably the **most important theory topic left in Unit 2**.

Examiners LOVE it because it represents how humans actually think.

---

### First Understand the Problem

Imagine I tell you:

Birds can fly.

You conclude:

Sparrow is a bird.

Therefore Sparrow can fly.

Correct ✅

---

Now I tell you:

Penguin is a bird.

You conclude:

Penguin can fly.

❌ Wrong.

Because:

Penguins cannot fly.

---

Question:

What happened?

Your old conclusion changed when new information arrived.

This is exactly:

## Non-Monotonic Reasoning

---

### Definition

Non-Monotonic Reasoning is a type of reasoning in which conclusions can be withdrawn or modified when new information becomes available.

Memorize this.

---

### Why "Non-Monotonic"?

First understand "Monotonic".

---

### Monotonic Reasoning

Once a conclusion is reached,  
it never changes.

Example:

All humans are mortal.

Ram is human.

Conclusion:

Ram is mortal.

No future information can change this.

---

## Non-Monotonic Reasoning

Conclusions may change.

Example:

Birds fly.

Conclusion:

Tweety flies.

New fact:

Tweety is a Penguin.

New conclusion:

Tweety cannot fly.

Old conclusion removed.

---

## Easy Human Example 🧐

You think:

Restaurant is open.

because:

It's Monday.

Then your friend says:

Today is a public holiday.

New conclusion:

Restaurant may be closed.

You changed your belief.

Humans do this all the time.

---

## Why is Non-Monotonic Reasoning Needed?

Real world changes constantly.

AI cannot assume all knowledge is permanent.

Examples:

### Medical Diagnosis

Fever  $\rightarrow$  Flu

New test:

Malaria detected

Diagnosis changes.

---

### Self Driving Cars

Road is clear.

New sensor input:

Pedestrian detected.

Decision changes.

---

### Stock Market

Stock will rise.

Breaking news:

Company bankruptcy.

Prediction changes.

---

## Example

Initial Knowledge:

$\text{Bird}(x) \rightarrow \text{Fly}(x)$

$\text{Bird}(\text{Tweety})$

AI concludes:

$\text{Fly}(\text{Tweety})$

---

New Information:

$\text{Penguin}(\text{Tweety})$

$\text{Penguin}(x) \rightarrow \neg \text{Fly}(x)$

New Conclusion:

$\neg \text{Fly}(\text{Tweety})$

Previous conclusion removed.

This is Non-Monotonic Reasoning.

---

## Types of Non-Monotonic Reasoning

Just remember names.

---

### 1. Default Reasoning ★

Most important.

Assume something is true unless proven otherwise.

Example:

Birds fly by default.

Until exception appears.

---

## 2. Circumscription

Assume only known facts are true.

Anything unknown is considered false.

---

## 3. Autoepistemic Reasoning

Reasoning about one's own knowledge.

Example:

I don't know that X is true.

---

## Default Reasoning Example

This comes in exams.

Default Rule:

Birds fly.

Given:

Tweety is Bird.

Conclusion:

Tweety flies.

---

New Fact:

Tweety is Penguin.

Conclusion changes:

Tweety cannot fly.

---

## Difference: Monotonic vs Non-Monotonic

★★★★★ Very Important

| Monotonic                               | Non-Monotonic                        |
|-----------------------------------------|--------------------------------------|
| Conclusions never change                | Conclusions may change               |
| New facts do not affect old conclusions | New facts can invalidate conclusions |
| Classical Logic                         | Real World AI                        |
| Fixed Knowledge                         | Dynamic Knowledge                    |

## Advantages

### 1. Realistic

Matches human thinking.

---

### 2. Handles Exceptions

Very important.

---

### 3. Adapts to New Information

Flexible.

---

### 4. Useful in Dynamic Environments

Robotics, diagnosis, planning.

---

# Disadvantages

## 1. More Complex

Harder than classical logic.

---

## 2. Computational Cost

Needs constant updates.

---

## 3. Possible Contradictions

New facts may conflict with old facts.

---

# Applications

## Expert Systems

Updating diagnoses.

---

## Robotics

Changing actions based on environment.

---

## Medical AI

Updating disease predictions.

---

## Planning Systems

Changing plans when new information arrives.

---

## 5 Marker Answer Structure

1. Definition
2. Bird-Penguin Example
3. Need for Non-Monotonic Reasoning
4. Types
5. Advantages
6. Applications

Done.

---

## Memory Hack

Think:

Old Belief

↓

New Information

↓

Belief Changes

That's literally the entire concept.

---

## Ultra Fast Revision

Monotonic:

Conclusion stays forever

Non-Monotonic:

Conclusion can change  
when new facts arrive

### One-line Example

Birds fly

Tweety is Bird

→ Tweety flies

Tweety is Penguin

→ Tweety cannot fly

🔥 Unit 2 Boss Fight Remaining:

## Statistical Reasoning

Very short topic, easy marks, and then Unit 2 is COMPLETE. 🚀📚😓

letsgoo

## Topic 10: Statistical Reasoning

🔥 Final topic of Unit 2.

Good news:

This is one of the easiest topics in the unit because it's mostly common sense.

---

## First Understand the Problem

Suppose I ask:

Will it rain tomorrow?

Can anyone say with 100% certainty?

❌ No.

But we can say:

80% chance of rain

based on past weather data.

This is:

## Statistical Reasoning

---

### Definition

Statistical Reasoning is the process of drawing conclusions and making decisions based on statistical data and probability.

---

### Simple Idea

Traditional Logic:

True

False

Statistical Reasoning:

Likely

Unlikely

80%

20%

---

### Example 1: Weather Forecast

Past Data:

100 cloudy days

80 rainy days

Reasoning:

$P(\text{Rain} \mid \text{Cloudy})$

=

$80/100$

$=$

0.8

Conclusion:

80% chance of rain

---

## Example 2: Medical Diagnosis

Data:

1000 patients

700 had fever

500 had flu

Doctor uses statistical reasoning:

Fever increases chance of flu

Not certainty.

Probability.

---

## Example 3: Cricket

Past Data:

Virat scored century

30 times in 100 matches

Prediction:

Probability of century

$= 30/100$

$= 0.3$

or

30%

---

## Why Statistical Reasoning is Needed?

Real-world information is uncertain.

Examples:

### Weather

Rain Prediction

---

### Stock Market

Price Prediction

---

### Healthcare

Disease Prediction

---

### Business

Customer Behavior

---

## Main Concepts

---

### 1. Probability

Most important.

Probability measures likelihood.

Formula:

$$P(E) = \frac{\text{Favourable Outcomes}}{\text{Total Outcomes}}$$

---

Example:

Throw a die.

Probability of getting 6:

1/6

---

## 2. Statistical Data

Information collected from observations.

Example:

Sales Data

Weather Data

Patient Data

---

## 3. Inference

Making conclusions from data.

Example:

Past 10 exams

8 times Unit 2 appeared

Inference:

Unit 2 likely important



# Statistical Reasoning Process

Very Important Flow.

Collect Data



Analyze Data



Calculate Probability



Draw Conclusion



Decision Making

---

## Statistical Reasoning vs Logical Reasoning

★★★★ Common theory question.

| Logical Reasoning | Statistical Reasoning |
|-------------------|-----------------------|
| Exact             | Probabilistic         |
| True/False        | Probability           |
| Certain facts     | Data-based estimates  |
| No uncertainty    | Handles uncertainty   |

## Example

Logical Reasoning:

All Humans are Mortal

Ram is Human

Therefore Ram is Mortal

100% sure.

---

Statistical Reasoning:

80% smokers get lung disease

Can we say:

Every smoker gets lung disease?

No.

Only probability.

---

## Applications

### Weather Forecasting

Rain prediction.

---

### Medical Diagnosis

Disease prediction.

---

### Finance

Risk analysis.

---

### Machine Learning

Model training.

---

### Market Analysis

Customer prediction.

---

# Advantages

## 1. Handles Uncertainty

Very useful.

---

## 2. Data Driven

Based on evidence.

---

## 3. Better Predictions

Useful in real-world systems.

---

## 4. Supports Decision Making

Used everywhere.

---

# Disadvantages

## 1. Not Always Accurate

Predictions may be wrong.

---

## 2. Requires Large Data

More data = better prediction.

---

## 3. Complex Calculations

Advanced statistics can be difficult.

---

## 5 Marker Answer Structure

1. Definition
  2. Need for Statistical Reasoning
  3. Probability Concept
  4. Process Diagram
  5. Applications
  6. Advantages
- 

## Quick Revision ⚡

Statistical Reasoning

=

Reasoning using  
Probability + Data

Examples:

Weather

Medical Diagnosis

Stock Market

Machine Learning

---

## UNIT 2 COMPLETE 🎉🔥

Ultimate Revision Sheet

Knowledge Representation

- Logic

- Semantic Networks
- Frames
- Production Rules

## KR Issues

- What to represent
- How to represent
- Frame Problem

## Predicate Logic

- Predicate
- Variables
- Constants
- $\forall$  and  $\exists$

## Facts / ISA / Instance

- Dog ISA Animal
- Tommy INSTANCE-OF Dog

## Resolution ★★★★★

- Convert clauses
- Negate goal
- Resolve
- Empty clause

## Procedural vs Declarative

- HOW vs WHAT

## Matching

- Find applicable rules

## Control Knowledge

- Choose best rule

## Symbolic Reasoning Under Uncertainty

- Probability
- Bayesian Networks
- Fuzzy Logic

## Non-Monotonic Reasoning ★★★★★

- Conclusions can change

## Statistical Reasoning

- Data + Probability
- 

## Exam Priority (Unit 2)

### 🔥 S-Tier

1. Resolution
2. Predicate Logic
3. Non-Monotonic Reasoning
4. Knowledge Representation

### ★ A-Tier

5. Frame Problem
6. ISA & Instance
7. Statistical Reasoning

### ✨ B-Tier

8. Matching
9. Control Knowledge

Unit 2 is now **exam-ready**. 🤔🚀

Next target should be **Unit 3**, because:

- Minimax

- Alpha-Beta Pruning
- NLP
- Explanation-Based Learning
- Neural Network Learning
- Genetic Learning

and past papers heavily favor those topics. 🔥

This is a very typical **Predicate Logic question**. Let's solve it exam-style. 🔥

---

## (i) Everyone likes Ram

Let:

Likes(x,y) = x likes y

Ram = constant

Statement:

Everyone likes Ram

Predicate Logic:

$\forall x Likes(x, Ram)$

Meaning:

For every person x,  
x likes Ram.

---

## (ii) No one is perfect

Let:

Perfect(x) = x is perfect

Statement:

No one is perfect

Predicate Logic:

$$\forall x \neg \text{Perfect}(x)$$

Alternative equivalent form:

$$\neg \exists x \text{Perfect}(x)$$

Meaning:

There does not exist any person who is perfect.

---

### (iii) Someone ate everything

Let:

Ate(x,y) = x ate y

Statement:

Someone ate everything

Predicate Logic:

$$\exists x \forall y \text{Ate}(x, y)$$

Meaning:

There exists a person x  
such that x ate every thing y.

---

### (iv) All basketball players are tall

Let:

BasketballPlayer(x)

Tall(x)

Statement:

All basketball players are tall

Predicate Logic:

$$\forall x [BasketballPlayer(x) \rightarrow Tall(x)]$$

Meaning:

For every x,  
if x is a basketball player,  
then x is tall.

---

## (b) Short Note: Handling Uncertainty Using Probabilistic Reasoning (6.5 Marks)

### Definition

Probabilistic reasoning is a technique used in AI to make decisions when information is uncertain, incomplete, or ambiguous. It uses probability values to represent the likelihood of events.

### Need

In real-world situations, AI rarely has complete information.

Examples:

- Medical diagnosis
- Weather forecasting
- Stock market prediction
- Spam email detection

### Probability Concept

Probability measures the chance of occurrence of an event.

$$P(E) = \frac{\text{Favourable Outcomes}}{\text{Total Outcomes}}$$

## Methods Used

### 1. Bayesian Networks

- o Represent uncertain knowledge using probabilities and graphs.

### 2. Bayes' Theorem

- o Updates probability when new evidence is received.

### 3. Fuzzy Logic

- o Handles partial truth values.

### 4. Statistical Reasoning

- o Draws conclusions from data.

## Advantages

- Handles uncertainty effectively.
- Supports better decision making.
- Works with incomplete information.
- Used in many real-world AI systems.

## Applications

- Medical expert systems
- Weather prediction
- Risk analysis
- Machine learning

## Conclusion

Probabilistic reasoning enables AI systems to make intelligent decisions under uncertainty by using probability theory and statistical methods.

---

## Exam Tip

For **6 marks**, write:

1. Definition
2. Need
3. Probability formula
4. Bayesian Networks
5. Applications
6. Advantages

That is usually enough for **full marks**. 😎🔥

This is a classic **Resolution Proof** question. Let's solve it exactly how you'd write it in the exam. 🔥

---

## Q5(a) Prove by Resolution that "Gurpreet buys a ticket"

### Step 1: Convert Statements into Predicate Logic

Let:

Enter(x) = x enters theatre

Buy(x) = x buys ticket

Money(x) = x has money

g = Gurpreet

#### Statement 1

"Everyone who enters a theatre has bought a ticket."

$\forall x [\text{Enter}(x) \rightarrow \text{Buy}(x)]$

Remove implication:

$\forall x [\neg \text{Enter}(x) \vee \text{Buy}(x)]$

Clause:

C1:  $\neg \text{Enter}(x) \vee \text{Buy}(x)$

---

## Statement 2

"Person who does not have money can't buy ticket."

$\forall x [\neg \text{Money}(x) \rightarrow \neg \text{Buy}(x)]$

Remove implication:

$\forall x [\text{Money}(x) \vee \neg \text{Buy}(x)]$

Clause:

C2:  $\text{Money}(x) \vee \neg \text{Buy}(x)$

(Note: This clause is actually extra information and won't be needed.)

---

## Statement 3

"Gurpreet enters a theatre."

$\text{Enter}(\text{Gurpreet})$

Clause:

C3:  $\text{Enter}(\text{Gurpreet})$

---

## Step 2: Negate Goal

Goal:

$\text{Buy}(\text{Gurpreet})$

Negation:

$\neg \text{Buy}(\text{Gurpreet})$

Clause:

C4:  $\neg \text{Buy}(\text{Gurpreet})$

---

## Step 3: Resolution

Resolve:

C1:  $\neg \text{Enter}(x) \vee \text{Buy}(x)$

C3:  $\text{Enter}(\text{Gurpreet})$

Substitute:

$x = \text{Gurpreet}$

Result:

C5:  $\text{Buy}(\text{Gurpreet})$

---

Now resolve:

C5:  $\text{Buy}(\text{Gurpreet})$

C4:  $\neg \text{Buy}(\text{Gurpreet})$

Result:

$\square$

(Empty Clause)

---

## Step 4: Conclusion

Since an **empty clause** ( $\square$ ) is obtained,  
a contradiction is found.

Therefore:

$\text{Buy}(\text{Gurpreet})$

is proved.

 **Hence, Gurpreet buys a ticket.**

---

## Q5(b) Difference Between Monotonic and Non-Monotonic Reasoning

| Monotonic Reasoning                     | Non-Monotonic Reasoning                  |
|-----------------------------------------|------------------------------------------|
| Conclusions never change                | Conclusions may change                   |
| New facts do not affect old conclusions | New facts can invalidate old conclusions |
| Used in classical logic                 | Used in real-world AI                    |
| Fixed knowledge                         | Dynamic knowledge                        |
| Example: Mathematics                    | Example: Medical diagnosis               |

### Example

Initial Knowledge:

Birds fly.

Tweety is a bird.

Conclusion:

Tweety flies.

New Fact:

Tweety is a penguin.

New Conclusion:

Tweety cannot fly.

This is Non-Monotonic Reasoning.

---

## Future of Non-Monotonic Reasoning in AI

1. Makes AI more human-like.
2. Helps AI adapt to changing information.
3. Important for expert systems and robotics.
4. Useful in self-driving cars and dynamic environments.

5. Plays a major role in modern intelligent decision-making systems.

## Conclusion

Non-monotonic reasoning is essential for building intelligent systems that can revise beliefs and make decisions in changing real-world situations.

---

## Exam Hack 🎯

For the resolution proof, the examiner mainly checks:

Clauses

Negated Goal

Resolution Steps

Empty Clause ( $\square$ )

If you show these clearly, you'll usually get full marks even if your notation isn't perfect. 🔥😓

This question is basically asking the **3 topics we just finished** 😎🔥

---

## Q4(a) Explain Resolution Principle with Example (6 Marks)

### Definition

Resolution is an inference rule used in Artificial Intelligence and Predicate Logic to derive conclusions by eliminating contradictory literals.

It works on the principle of **proof by contradiction**.

---

### Resolution Principle

Steps:

1. Convert statements into clauses.
  2. Negate the goal.
  3. Apply resolution.
  4. Obtain empty clause ( $\square$ ).
  5. Goal is proved.
- 

## Example

Given:

All humans are mortal.  
Ram is a human.

Prove:

Ram is mortal.

## Predicate Logic

$\forall x [\text{Human}(x) \rightarrow \text{Mortal}(x)]$

$\text{Human}(\text{Ram})$

## Clause Form

C1:  $\neg \text{Human}(x) \vee \text{Mortal}(x)$

C2:  $\text{Human}(\text{Ram})$

Negated Goal:

C3:  $\neg \text{Mortal}(\text{Ram})$

---

## Resolution

Resolve C1 and C2:

$\text{Mortal}(\text{Ram})$

Resolve with C3:

$\text{Mortal}(\text{Ram})$

$\neg \text{Mortal}(\text{Ram})$

Result:

□

(Empty Clause)

Hence:

Ram is Mortal

is proved.

---

## Q4(b)(i) Symbolic Reasoning

### Definition

Symbolic Reasoning is the process of representing knowledge using symbols and applying logical rules to derive conclusions.

---

### Example

Knowledge:

Human(Ram)

$\forall x [\text{Human}(x) \rightarrow \text{Mortal}(x)]$

Reasoning:

Mortal(Ram)

---

### Applications

- Expert Systems
  - Knowledge Representation
  - Theorem Proving
-

## Q4(b)(ii) Non-Monotonic Reasoning

### Definition

Non-Monotonic Reasoning is a reasoning process in which conclusions can be changed or withdrawn when new information becomes available.

---

### Example

Initial knowledge:

Birds fly.

Tweety is a bird.

Conclusion:

Tweety flies.

New fact:

Tweety is a penguin.

New conclusion:

Tweety cannot fly.

Old conclusion changed.

---

### Applications

- Robotics
  - Medical Diagnosis
  - Expert Systems
-

# Q4(b)(iii) Statistical Reasoning

## Definition

Statistical Reasoning is the process of making decisions and drawing conclusions based on statistical data and probability.

---

## Example

Past Data:

100 cloudy days

80 rainy days

Probability:

$P(\text{Rain} | \text{Cloudy})$

$= 80/100$

$= 0.8$

Conclusion:

80% chance of rain

---

## Applications

- Weather Forecasting
  - Risk Analysis
  - Machine Learning
  - Medical Prediction
-

# 30-Second Revision Before Exam ⚡

## Resolution

Clauses

↓

Negate Goal

↓

Resolve

↓

Empty Clause ( $\square$ )

## Symbolic Reasoning

Knowledge represented as symbols

+

Logical Rules

## Non-Monotonic

New Information

↓

Conclusion Changes

## Statistical Reasoning

Data

+

Probability

=

Decision

This exact answer format is more than enough for full marks in this PYQ. 🔥📚😓

These are classic **short notes / differentiation questions**. Easy marks. 🔥

---

## Q5(a) Differentiate between Inductive and Deductive Reasoning

| Inductive Reasoning        | Deductive Reasoning                          |
|----------------------------|----------------------------------------------|
| Specific → General         | General → Specific                           |
| Based on observations      | Based on rules/facts                         |
| Conclusion may be probable | Conclusion is certain (if premises are true) |
| Used to discover rules     | Used to apply rules                          |
| Bottom-up approach         | Top-down approach                            |

### Example of Inductive Reasoning

Observe:

Crow 1 is black

Crow 2 is black

Crow 3 is black

Conclusion:

All crows are black

(May be true, not guaranteed)

---

### Example of Deductive Reasoning

Given:

All humans are mortal.

Ram is a human.

Conclusion:

Ram is mortal.

(Guaranteed true)

---

## Q5(b) Difference Between Forward Chaining and Backward Chaining

★★★★ Very Important

### Forward Chaining

Starts from facts and moves toward conclusion.

#### Example

Facts:

Fever  
Cough

Rule:

IF Fever AND Cough  
THEN Flu

Conclusion:

Flu

Flow:

Facts  
↓  
Rules  
↓  
Goal

---

### Backward Chaining

Starts from goal and checks whether facts support it.

#### Example

Goal:

Flu?

System asks:

Do you have fever?  
Do you have cough?

If yes:

Flu

Flow:

Goal

↓

Rules

↓

Facts

---

## Comparison Table

| Forward Chaining                       | Backward Chaining           |
|----------------------------------------|-----------------------------|
| Data-driven                            | Goal-driven                 |
| Starts with facts                      | Starts with goal            |
| Moves forward                          | Moves backward              |
| Suitable when many facts are available | Suitable when goal is known |
| Used in monitoring systems             | Used in diagnosis systems   |

## Example

Forward:

Fever + Cough

↓

Flu

Backward:

Flu?

↓

Check Fever

Check Cough

---

## Q5(c) Define Propositional Logic

### Definition

Propositional Logic is a branch of logic in which statements (propositions) are represented as either True or False and combined using logical operators.

---

### Logical Operators

| Symbol        | Meaning |
|---------------|---------|
| $\wedge$      | AND     |
| $\vee$        | OR      |
| $\neg$        | NOT     |
| $\rightarrow$ | IMPLIES |

### Example

Let:

P = It is raining

Q = Road is wet

Statement:

$P \rightarrow Q$

Meaning:

If it is raining,  
then the road is wet.

---

### Advantages

- Simple to understand
- Easy implementation

- Used in reasoning systems
- 

## Limitation

Cannot represent relationships between objects.

Example:

Ram is a student

Shyam is a student

Predicate Logic handles this better.

---

## Exam Revision Sheet ⚡

### Inductive

Specific → General

Examples → Rule

### Deductive

General → Specific

Rule → Conclusion

### Forward Chaining

Facts → Goal

### Backward Chaining

Goal → Facts

### Propositional Logic

True / False Statements

+

Logical Operators

These three answers together are easily worth **10–12 marks** if they appear. 🔥😎